■ 1.3.3 Getting Information from *Mathematica*

You will often need to look up information on *Mathematica* functions that you want to use. In more sophisticated *Mathematica* front ends, there are usually graphical ways to request information on a particular object. In all cases, however, you can use what is discussed in this section to get information directly from the *Mathematica* kernel.

?*Name*	show information on *Name*
??*Name*	show extra information on *Name*
?*Aaaa**	show information on all objects whose names begin with *Aaaa*
?++ etc.	get information on special input forms

Ways to get information on *Mathematica* objects.

This gives information on the built-in function `Log`.

```
In[1]:= ?Log

Log[(base:E), x] gives the logarithm of x (default is
  natural log).
```

This gives some extra information on `Log`. `Attributes` will be discussed in Section 2.2.18.

```
In[1]:= ??Log

Log[(base:E), x] gives the logarithm of x (default is
  natural log).
Attributes[Log] = {Listable, Protected}
```

This gives information on all *Mathematica* objects whose names begin with `L`. When there is more than one object, *Mathematica* just gives you the name of each of them.

```
In[1]:= ?L*

LCM              LerchPhi         ListContourPlot
Label            Less             ListDensityPlot
LaguerreL        LessEqual        ListPlot
Last             Level            ListPlot3D
LatticeReduce    LightSources     Listable
LeafCount        Lighting         Literal
Left             Limit            Locked
LegendreP        Line             Log
LegendreQ        LineBreak        LogIntegral
LegendreType     LinearSolve      LogicalExpand
Length           List
```

?*Aaaa* will give you information on the particular object whose name you specify. Using the "metacharacter" `*`, however, you can get information on collections of objects with similar names. The rule is that `*` is a "wild card" which can stand for any sequence of ordinary characters. So, for example, `?L*` gets information on all objects whose names consist of the letter `L`, followed by any sequence of characters.

You can put a * anywhere in the string you ask ? about. For example, ?*Expand would give you all objects whose names *end* with Expand. ?x*0 would give you objects whose names start with x, and end with 0, and have any sequence of characters in between. (You may notice that the way you use * to specify names in *Mathematica* is similar to the way you use * in UNIX and other operating systems to specify file names.)

You can ask for information on most of the special input forms that *Mathematica* uses. This asks for information about the := operator.

```
In[1]:= ?:=

a:=b or SetDelayed[a,b] defines that the value of a is
  always the current value of b (delayed assignment).
```

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; http://www.wolfram.com/; 1-800-441-6284.